

Medicare Healthcare Utilization & Spending Analysis

Executive Summary

This project analyzed Medicare beneficiary enrollment, inpatient claims, outpatient claims, and prescription drug event data to identify healthcare utilization patterns, spending trends, and high-cost beneficiary populations. Data was imported into PostgreSQL, transformed using SQL views, and visualized in Power BI through an interactive dashboard. The analysis revealed that outpatient services accounted for the largest share of healthcare expenditures, while a relatively small subset of beneficiaries generated disproportionately high healthcare costs. Demographic analysis further highlighted differences in spending and utilization across age groups and beneficiary populations.

Project Overview

The objective of this project was to examine Medicare healthcare utilization and spending patterns using publicly available claims data. The analysis focused on four key areas:

- Healthcare spending by service type
- Beneficiary demographic characteristics
- High-cost beneficiary identification
- Spending distribution and utilization trends

The project demonstrates the use of SQL, PostgreSQL, data modeling, data transformation, and Power BI visualization techniques commonly used in healthcare analytics environments.

Data Sources

The analysis utilized Medicare synthetic beneficiary and claims data, including:

- Beneficiary Enrollment Data
- Inpatient Claims
- Outpatient Claims
- Part D Prescription Drug Events

The dataset contained approximately 10,000 Medicare beneficiaries and over one million healthcare service events across multiple service categories.

Methodology

Data was imported into PostgreSQL and transformed using SQL queries and analytical views. Multiple views were created to aggregate healthcare spending, beneficiary demographics, and utilization metrics.

The transformed data was connected to Power BI, where interactive dashboards were developed to analyze:

- Total healthcare spending
- Service utilization patterns
- Beneficiary demographics
- High-cost beneficiary populations
- Spending concentration and distribution

Key metrics included total spending, average spending per beneficiary, service event counts, demographic distributions, and spending by healthcare service type.

Key Findings

Healthcare Spending

Healthcare spending exceeded \$2.25 billion across all analyzed service categories.

Outpatient services represented the largest share of healthcare expenditures, accounting for more than 60% of total spending. Inpatient services generated significantly higher costs per encounter despite lower utilization volume, while prescription drug spending represented a comparatively small share of overall expenditures.

Beneficiary Demographics

Analysis of beneficiary demographics revealed a balanced gender distribution and a diverse beneficiary population. Beneficiaries under age 65 represented the largest enrollment group, reflecting the inclusion of disabled Medicare beneficiaries alongside traditional age-based enrollment populations.

High-Cost Beneficiaries

Healthcare spending was highly concentrated among a relatively small number of beneficiaries. The highest-cost beneficiary generated approximately \$49 million in healthcare expenditures, demonstrating the substantial impact that a small number of complex cases can have on overall healthcare spending.

Average spending among beneficiaries with claims was significantly lower than the highest observed expenditures, highlighting considerable variation in healthcare utilization across the population.

Spending Distribution

Spending patterns indicated that a limited subset of beneficiaries accounted for a disproportionate share of overall healthcare costs. This distribution is consistent with healthcare spending trends observed in real-world populations, where high-cost patients often drive a significant percentage of total expenditures.

Business Implications

The findings demonstrate the importance of identifying high-cost populations and understanding healthcare utilization patterns. Organizations can use similar analyses to:

- Identify cost drivers
- Monitor healthcare utilization trends
- Target care management interventions
- Improve resource allocation
- Support population health initiatives

Understanding spending concentration and utilization patterns can help healthcare organizations focus resources on populations with the greatest opportunity for improved outcomes and cost management.

Technical Skills Demonstrated

Data Analytics

- Data Cleaning
- Data Transformation
- Exploratory Data Analysis
- Healthcare Analytics
- Business Intelligence

SQL & Database Technologies

- PostgreSQL
- SQL Queries
- Aggregate Functions
- Joins
- Views
- Data Modeling

Data Visualization

- Power BI
- Dashboard Design
- KPI Development
- Interactive Reporting
- Data Storytelling

Conclusion

This project successfully analyzed Medicare healthcare utilization and spending patterns through the integration of PostgreSQL and Power BI. The analysis identified key spending drivers, demographic trends, and high-cost beneficiary populations while demonstrating practical healthcare analytics techniques used in real-world business and healthcare environments.